Hao Zhang

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Education

University of Glasgow

- Ph.D. in Mathematics, December 2025
- Supervisor: Prof. Michael Wemyss
- Thesis: *Local Forms for the Double A_n Quiver and Gopakumar–Vafa Invariants*

Nankai University

- M.Sc. in Mathematics
- Supervisor: Prof. Ming Ding
- Dissertation: *Ring-theoretic Properties of Generalized Cluster Algebras*

Beijing University of Posts and Telecommunications

- B.Eng. in Computer Science and Technology

Research Interests

My research lies at the intersection of algebraic geometry and representation theory, focusing on noncommutative methods in birational geometry and their connections with enumerative invariants and Homological Mirror Symmetry.

- **Crepant resolutions of compound Du Val (cDV) singularities.** Classification via contraction algebras and quiver potentials.
- **Curve-counting invariants.** Generalised Gopakumar–Vafa invariants for crepant partial resolutions.
- **Noncommutative crepant resolutions (NCCRs).** Connections with Homological Mirror Symmetry and derived deformation theory.
- **A_∞-structures of contraction algebras.** Homotopical methods for configurations of exceptional curves.

Broadly, I develop noncommutative approaches to singularity theory, using contraction algebras and NCCRs to study birational geometry and enumerative invariants.

Publications and Preprints

1. **H. Zhang**, *Local Forms for the Double A_n Quiver. Mathematische Zeitschrift*, to appear. arXiv:2412.10042.
2. **H. Zhang**, *Gopakumar–Vafa Invariants Associated to cA_n Singularities.* arXiv:2504.03139.

[1]Updated March 5, 2026

Talks and Presentations

Research Talks

- *Local Forms for the Double A_n Quiver and Gopakumar–Vafa Invariants*, Yau Mathematical Sciences Center, Tsinghua University, 2025
- *Local Forms for the Double A_n Quiver and Gopakumar–Vafa Invariants*, University of Warwick, 2025
- *Local Forms for the Double A_n Quiver and Gopakumar–Vafa Invariants*, University of Edinburgh, 2025
- *Gopakumar–Vafa Invariants Associated to cA_n Singularities*, University of Glasgow, 2024
- *Local Forms for the Double A_n Quiver*, University of Glasgow, 2023

Poster Presentations

- *Local Forms for the Double A_n Quiver and Gopakumar–Vafa Invariants*, Geometry, Higher structures, and Physics, Edinburgh, 2025
- *Local Forms for the Double A_n Quiver and Gopakumar–Vafa Invariants*, Advances in Representation Theory of Algebras, Cologne, 2025
- *Local Forms for the Double A_n Quiver and Gopakumar–Vafa Invariants*, Summer Research Institute in Algebraic Geometry, Colorado State University, 2025

Teaching Experience

1. Tutor for undergraduate mathematics, University of Glasgow, 2025
2. Marker for undergraduate assignments, Nankai University, 2020

Selected Conferences Attended

1. Geometry, Higher structures, and Physics, Edinburgh, 2025
2. New Developments in Singularity Theory, Miami, 2025
3. Advances in Representation Theory of Algebras, Cologne, 2025
4. Representations, Moduli and Duality, Lausanne, 2025
5. Summer Research Institute in Algebraic Geometry, Colorado State University, 2025
6. Winter School, Lancaster, 2024
7. Computational Algebraic Geometry Workshop, Durham, 2024
8. Recent Developments in Noncommutative Algebraic Geometry, Berkeley, 2024
9. Winter School, Warwick, 2023
10. Derived Categories, Moduli Spaces, and Counting Invariants, London, 2023
11. Quivers, Clusters, Moduli and Stability, Oxford, 2023
12. Noncommutative Shapes, Antwerp, 2022
13. E-RNG Glasgow Meeting, Glasgow, 2022